UNIT 2: INTRODUCTION TO PYTHON

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Unit - 2	**Introduction:** Features of Python, execution of Python programs, Python virtual machines, memory management, garbage collection, comparison between C and Python. **Data Types:** Comments, docstrings, built-in data types, strings, sets, literals, user-defined data types, constants, identifiers, reserved words and naming conventions in python. **Operators:** Arithmetic, assignment, unary, relational, logical, Boolean, bitwise, membership, identity operators, operator precedence and associativity. **Control statements:** if, if-else, if-elif else, while, for, nested loops, break, continue, pass, assert and return statements	**12**

2. Dr. R. Nageswara Rao; "Core Python Programming", Dreamtech press, Third edition, 2018.

INTRODUCTION TO PYTHON

- **Python** is a **programming language** that **combines features of C and Java**
- **C**: elegant style of developing programs
- **Java**: Classes and objects like Java
- Python will have **very few lines of code compared to C and Java**.

- **Example: Python Program to add two numbers**

```
a=b=10  # take two variables and store 10 into them
print("Sum= ",(a+b)) #display their sum
```

- Few lines of code and easy to understand
- Developed by Guido Van Rossum in 1991
- Open source software: anybody can freely download it

INTRODUCTION TO PYTHON : FEATURES OF PYTHON

- **Simple**: Python is a **simple** programming language. When we read a python program we feel like **reading English sentences**. Hence **developing and understanding** is easier.
- **Easy to learn**: Uses very few keywords. **Simple structure. Resembles C language**. Migrating from C to python is easy.
- **Open source**: **No need to pay** for python software. Python can be freely downloaded from www.python.org website.
- **High level language**: Programming languages are of two types: low level and high level
High level languages use English words to develop programs. **Python is high level language.**
- **Dynamically typed:** we **do not declare anything**. If a name is assigned to an object of one type, it may later be assigned to an object of different type.
- **Platform independent**: Python programs **are not dependent on any specific operating system**. Python can be used on almost all operating systems like UNIX,Linux, Windows,Mac.
- **Portable**: Python programs **yields the same result on any computer** in the world as they are platform independent
- **Procedure and object oriented**: procedure oriented uses functions and procedures(eg: C)
object oriented: uses classes and objects(eg: C++ and Java)

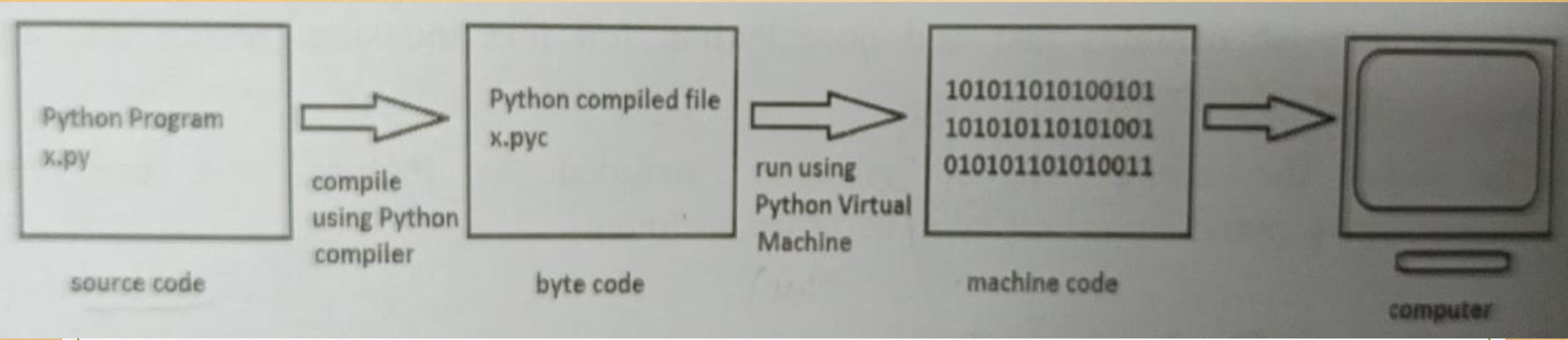
INTRODUCTION TO PYTHON : FEATURES OF PYTHON

- **Interpreted**: Python program is called source code. **After writing python program we should compile the source code using python compiler.**
- **Extensible**: Programs **written in C and C++ can be integrated into python and executed using PVM(**Python virtual machine)
- **Embeddable**: We **can insert python programs into C or C++ programs**.
- **Huge library:** Python has a **big library** which can be used on any operating systems
- **Database connectivity**: Python uses **interfaces to connect its programs to all major databases**.
- **Scalable**: Python programs are **scalable since they can run on any platform** and use features of new platform
- **Batteries included:** Huge library of python **contains several small applications** which are already developed and immediately available for programmers.

  numpy : is a package for processing arrays

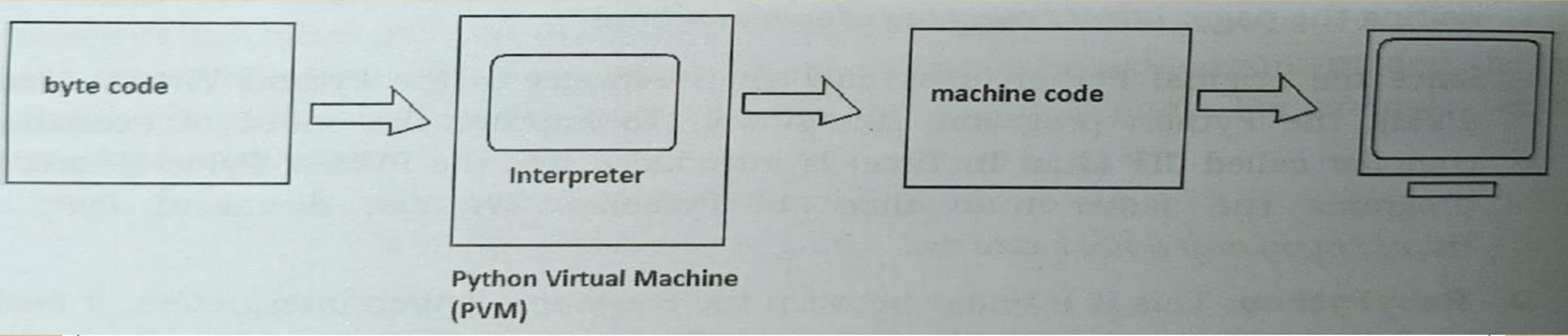
  w3lib: is a package of web related functions

EXECUTION OF A PYTHON PROGRAM



- We write a python program(source code) with the **name x.py**
- **Compiler** converts python program into **byte code**
- **Byte code** instructions are contained in the **file x.pyc**
- **PVM** converts **byte code to machine code(0s and 1s) using interpreter**.
- **Machine code instructions are executed by processor** and **results displayed**.
- **Interpreter** translates the program source **line by line and hence slow**.
- To **rectify** this problem sometimes a **Just in time compiler** is added to **improve the speed**.
- **Source code→byte code→machine code→ output** are **internally completed** and then the final result is displayed.

PYTHON VIRTUAL MACHINE



- Program **source code→ byte code→ machine code**
- **Byte code fixed set of instructions representing all operations**. Size of each instruction is 1 byte.
- **Byte code** instructions are in **.pyc file.**
- Python virtual machine(**PVM)** converts **byte code to machine code**
- **PVM** uses **interpreter** to **convert byte code to machine code** and sends machine code to computer processor for execution.

Memory management in python

- In C the programmer should allocate or deallocate memory dynamically(using functions like malloc( ) , free( ).
- In **python memory allocation and deallocation are done automatically by python virtual machine(PVM).**
- Everything is considered as an object in python.eg: strings , functions are objects.
- Memory manager inside **PVM allocates memory required for objects** created in a python program.
- **Objects are stored in heap.**
- **Size** of the **heap depends** on the **random access memory(RAM)** of our computer and can increase/decrease based on requirement of the program.

Memory management in python

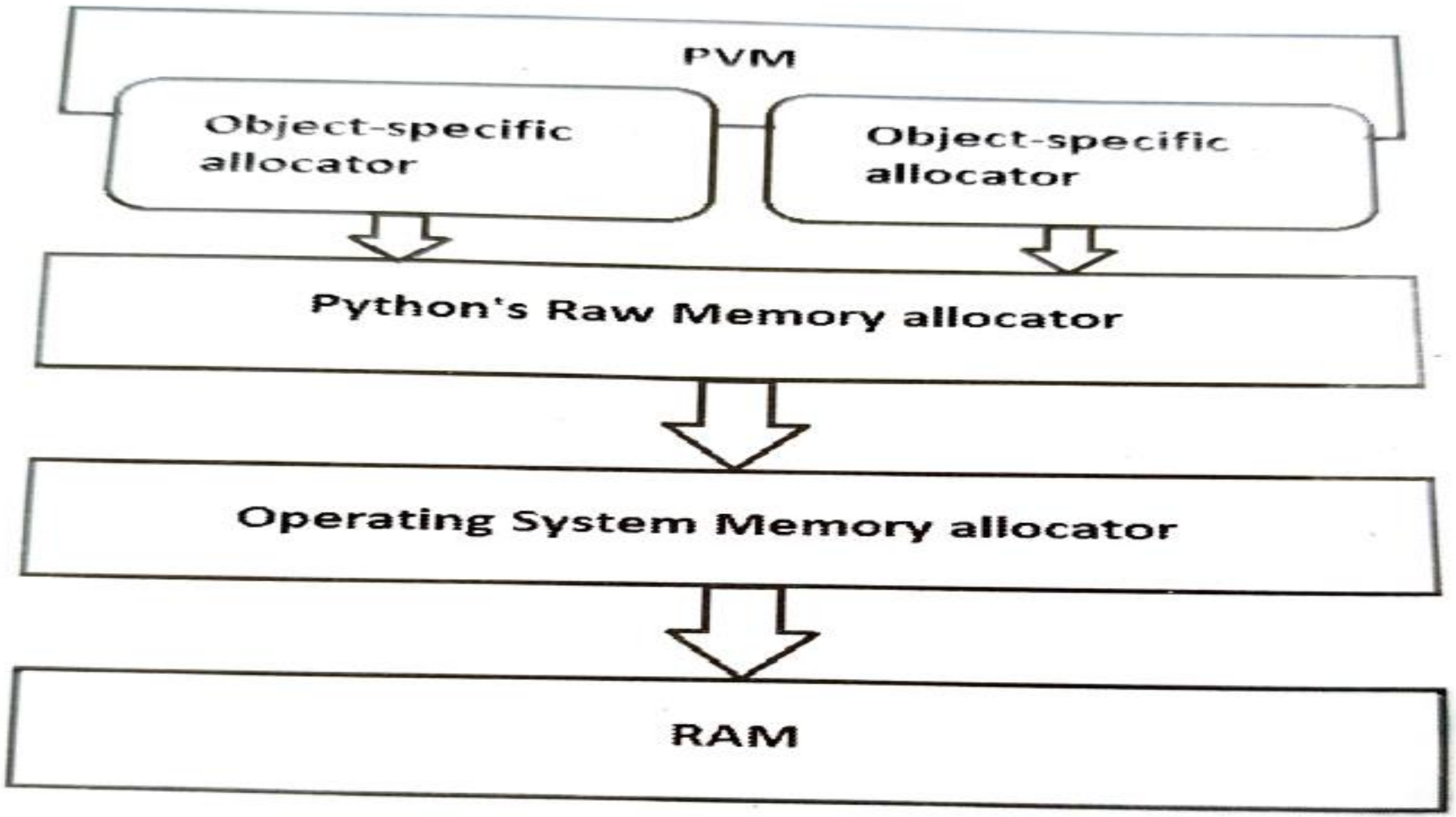


Figure 1.5: Allocation of memory by Python's Virtual Machine (PVM)

- **Operating system:** Memory(**RAM**) for any program is **allocated by the operating system**.
- **Raw memory allocator:** oversees whether **enough memory is available** to it for storing objects.
- **Object specific allocators: implement** different types of **memory management policies depending on the type of objects.** E.g. integer number should be stored in memory in one way and a string should be stored in a different way.

Garbage collection in Python

- **Garbage collector is a module** in python that is useful to **delete objects from memory, which are not used in the program.**
- The **module that represents the garbage collector is named as gc**.
- Garbage collector is the **simplest way to maintain count for each object regarding how many times the object is referenced(or used).**
- When **object has some count, garbage collector will not remove it from memory**.
- When an **object is found with reference count 0, it will be deleted from the memory**.
- Garbage collector **can detect reference cycles and remove objects in the cycle**.

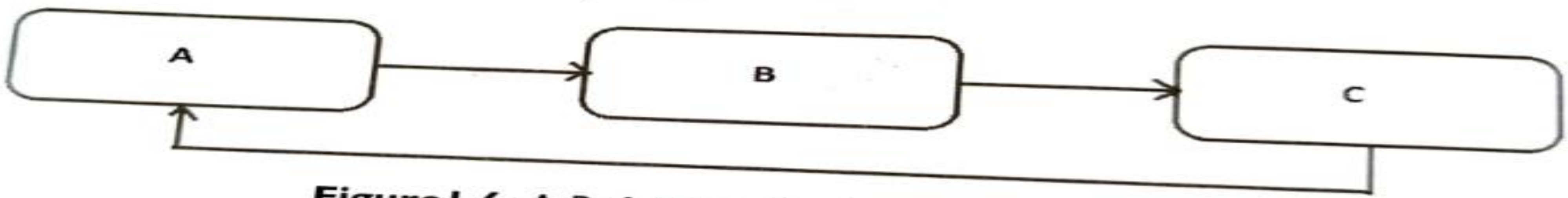


Figure1.6: A Reference Cycle of Three Objects

Garbage collection in Python

- **Garbage collector classifies object into three generations**.
- **Newly created objects** are considered as **generation 0 objects**.
- When garbage collector examines the object in memory and **does remove an object from memory due(because it is used by a program)** then that **object is placed into the next generation that is generation 1.**
- When garbage **collector intends to delete** the objects for the **second time and the object survives for the second time, then it is placed in generation 2.**
- **Older objects** belongs to **generation 2**.
- Garbage collector tries to **delete younger objects** which are not referenced in the program rather than the old objects.
- **Garbage collector run automatically**. Python schedules garbage collector depending on a number called as threshold.
- In some cases where **reference cycles are found in the program, it is better to run the garbage collector manually.**

COMPARISON BETWEEN C AND PYTHON

C	PYTHON
C is a procedure oriented programming language. It does not contain features like classes , objects , inheritance etc.	Python is object oriented programming language. It contains features like classes objects inheritance, polymorphism etc.
C programs execute faster	Slower as compared to C
It is compulsory to declare the datatypes of variables arrays etc in C	Type declaration is not required in python
C language type discipline is static and weak	Python type discipline is dynamic and strong
Pointers concept is available in C	Python does not use pointers.
C does not have exception handling	Python handles exceptions and hence it is robust
C has do while, while and for loops	Python has while and for loops

COMPARISON BETWEEN C AND PYTHON

C	PYTHON
C has switch statement	Python does not have switch statements
The variable in for loop does dot increment automatically.	Variable in for loop increments automatically
Programmer should allocate and deallocate memory using malloc( ), calloc( ), free( ) and realloc( )	Memory allocation and deallocation done by PVM
C supports single and multidimentional arrays	Python supports only single dimentional arrays. To work with multidimentional arrays we should use third part applications like numpy
Array index should be a positive integer	Array index can be positive or negative integer.Negative index represents locations from the end of the array.
Indentation of statements is not necessary in C	Indentation is required to represent a block of statements
Semicolon is used to terminate the statements	New line indicates end of statements

COMPARISON BETWEEN JAVA AND PYTHON

Java	Python
Java programs are verbose. It means they contain more number of lines.	Python programs are concise and compact. A big program can be written using very less number of lines.
It is compulsory to declare the datatypes of variables, arrays etc. in Java.	Type declaration is not required in Python.
Java language type discipline is static and weak.	Python type discipline is dynamic and strong.

Java	Python
Java has do... while, while, for and for each loops.	Python has while and for loops.
Java has switch statement.	Python does not have switch statement.
The variable in for loop does not increment automatically. But in for each loop, it will increment automatically.	The variable in the for loop increments automatically.
Memory allocation and deal location is done automatically by JVM (Java Virtual Machine).	Memory allocation and deal location is done automatically by PVM (Python Virtual Machine).
Java supports single and multi-dimensional arrays.	Python supports only single dimensional arrays. To work with multi-dimensional arrays, we should use third party applications like numpy.
The array index should be a positive integer.	Array index can be positive or negative integer number. Negative index represents locations from the end of the array.

of an array is not supp...	
Indentation of statements is not necessary in Java.	Indentation is required to represent a block of statements.
A semicolon is used to terminate the statements and comma is used to separate expressions.	New line indicates end of the statements and semicolon is used as an expression separator.
In Java, the collection objects like Stack, LinkedList or Vector store only objects but not primitive datatypes like integer numbers.	Python collection objects like lists and dictionaries can store objects of any type, including numbers and lists.

THANK YOU☺

by PRACHI SURLAKAR